

# M.W.P. DEWMI ABEYKOON

Data Science Undergraduate | Machine Learning & AI Enthusiast

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[Portfolio](#) | [LinkedIn](#) | [GitHub](#)

## SUMMARY

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Final-year Data Science undergraduate with a strong foundation in mathematics and a keen interest in Artificial Intelligence, Machine Learning, and Data Analytics. Experienced in building end-to-end ML pipelines, explainable AI systems, and full-stack prototypes through hands-on academic and personal projects. Curious, adaptable, and committed to continuous learning, with a strong analytical mindset and problem-solving approach. Eager to apply academic knowledge to real-world challenges while growing professionally in the field of Data Science.

## TECHNICAL SKILLS

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**Data Science & Machine Learning:** Machine Learning, Deep Learning (TensorFlow), NLP, Predictive Modeling, Scikit-learn, Data Analysis, MLOps, Model Deployment, Model Evaluation, Power BI

**Big Data:** Apache Spark, PySpark

**Programming Languages:** Python, SQL, Java, JavaScript, Kotlin

**BI & Data Visualization:** Power BI, Streamlit

**Databases & Cloud:** MongoDB, Firebase

**Development & Platforms:** React.js, Android Studio, Docker (Basic), Linux, Git/GitHub

## PROJECTS

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### DeepShield — Explainable Multimodal Framework for Fake News Detection

Final Year Research Project — Ongoing | Python, PyTorch, CLIP, BERT, RAG, FAISS, FastAPI, React.js

- Designing a multimodal fake news detection framework that fuses pretrained image (CLIP) and text (BERT) models with metadata analysis and RAG-based evidence verification to assess news credibility.
- Applying Explainable AI techniques (Grad-CAM, SHAP) to generate an interpretable Trust Score and human-readable justifications, moving beyond traditional binary classification.
- Building a web-based prototype (FastAPI + React.js) evaluated on Fakeddit, FakeNewsNet, and LIAR datasets using Accuracy, Precision, Recall, F1-score, and ROC-AUC.
- Co-authoring research under SLTC Research University, Department of Software Engineering.

### Kamatha — AI-Powered Plant Disease Detection App

Team Project | TensorFlow, CNN, JavaScript, Gemini API

- Contributed to the development of an AI-powered mobile application enabling real-time plant disease detection from leaf imagery using a CNN model.
- Integrated the Gemini API to deliver AI-generated diagnostics and natural/chemical treatment recommendations, alongside an interactive chatbot assistant.
- Implemented multi-language support and persistent chat history to enhance usability across diverse user bases.

**Project:** [github.com/dewmiabeykoon/kamatha-plant-disease-detection](https://github.com/dewmiabeykoon/kamatha-plant-disease-detection)

**Live Site:** [kamathaweb.netlify.app](https://kamathaweb.netlify.app) | [kamathaapp.netlify.app](https://kamathaapp.netlify.app)

### Customer Churn Prediction — End-to-End MLOps Pipeline

Python, Machine Learning, MLOps, Docker, MLflow

- Architected and deployed an end-to-end machine learning pipeline for customer churn prediction, spanning data preprocessing, feature engineering, model training, and deployment.
- Applied MLOps best practices, including basic Docker containerization, to ensure model reproducibility, version control, and streamlined retraining workflows.

**GitHub:** [github.com/dewmiabeykoon/churn-prediction-mlops](https://github.com/dewmiabeykoon/churn-prediction-mlops)

### Supermarket Business Intelligence — Data Analytics Case Study

*Python, SQL, Power BI*

- Executed end-to-end analysis of a large-scale supermarket sales dataset, including data cleaning, preprocessing, and statistical hypothesis testing.
- Trained and evaluated machine learning models to extract actionable business insights from sales and customer behavior trends.
- Engineered an interactive Power BI dashboard translating analytical findings into decision-ready visualizations for stakeholders.

**GitHub:** [github.com/dewmiabeykoon/supermarket-bi-case-study](https://github.com/dewmiabeykoon/supermarket-bi-case-study)

### **LankaSmartMart — Android Shopping Application**

*Kotlin, Android Studio, Firebase*

- Engineered a full-featured Android e-commerce application with secure user authentication and real-time data synchronization via Firebase.
- Designed a scalable app architecture supporting product browsing, cart management, and end-to-end purchase workflows.

**GitHub:** [github.com/dewmiabeykoon/LankaSmartMart](https://github.com/dewmiabeykoon/LankaSmartMart)

### **Student Management System**

*Java*

- Developed a Java-based application implementing structured CRUD logic for efficient management of student records.

**GitHub:** [github.com/dewmiabeykoon/Student-management-app](https://github.com/dewmiabeykoon/Student-management-app)

## **EDUCATIONAL QUALIFICATIONS**

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### **BSc (Hons) in Data Science — Undergraduate, Final Year (2023 – 2027)**

SLTC Research University

### **G.C.E. Advanced Level — Physical Science Stream (2022)**

Visakha Girls' High School, Badulla

Combined Mathematics, Physics, Chemistry

### **G.C.E. Ordinary Level (2018)**

Visakha Girls' High School, Badulla

## **CERTIFICATIONS & ACHIEVEMENTS**

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- Introductory SQL — Kaggle (June 2026)
- Advanced SQL — Kaggle (June 2026)
- Certificate of Comprehensive UI/UX Design — FSD Academy (August 2025) — completed 7 projects spanning mobile app, full website, and client-based UX process flow work
- Certificate of Participation, "Will AI Replace Your Business or Transform It?" Webinar — Sri Lanka Technology Campus (June 2026)
- Certificate of Appreciation, Codemania v5.0 (12-Hour Algorithmic Coding Competition), Team Pixel Mind — IEEE Computer Society Student Branch Chapter, SLTC (February 2025)

## **SOFT SKILLS & LANGUAGES**

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**Professional Skills:** Critical Thinking, Analytical Thinking, Research, Adaptability, Teamwork, Communication

**Languages:** Sinhala (Native), English (Fluent)

## **REFERENCES**

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*Available upon request.*